

Hashan Fernando

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Research Interests

Machine Learning, Explainable AI, Topological Data Analysis, Epidemiology

Education

Haslam College of Business , University of Tennessee, Knoxville, TN M.S. Statistics GPA.: 4.00 out of 4. Advisor: Dr. Adam Spannaus	2022 – 2026
Postgraduate Institute of Science , University of Peradeniya, Sri Lanka M.S. Data Science GPA: 3.75 out of 4.	2020 – 2022
Faculty of Science University of Peradeniya, Sri Lanka B.Sc. Statistics and Operations Research GPA: 3.06 out of 4.	2015 – 2018

Experience

Graduate Research Assistant (Ph.D. Student) Advanced Computing in Health Sciences Section, Oak Ridge National Laboratory , Oak Ridge, TN	Aug 2022 – Apr 2026
Designed a novel topological data analysis-Bayesian spatial algorithm to identify complex spatial patterns in opioid overdose risk in the USA, mitigating risk estimate uncertainty in the baseline Bayesian spatial models	
Graduate Research Assistant Department of Statistics and Computer Science, University of Peradeniya , Sri Lanka	Dec 2019 – Feb 2022
Developed a smart seal and audio detection system for protecting steel-based cargo using IoT	
Data Scientist Intern VizuaMatix , Colombo, Sri Lanka	Jan 2019 – Jul 2019
- Developed a machine learning model to predict non-performing customers in a banking system - Developed a route optimization model for food distribution trucks to improve the efficiency and reduce operational costs	
Planning Intern MAS Holdings , Biyagama, Sri Lanka	Oct 2018 – Jan 2019
Developed multiple VBA application for planning inventory	

Publications

- H. Fernando**, "Unmeasured Spatial Risk in Disease Modeling: A Topological Data Analysis Approach", *Manuscript in preparation*, 2026
- A. Deas, **H. Fernando**, A. Heidi, A. Kapadia, J. Trafton, A. Spannaus, V. Maroulas, "Identifying spatiotemporal patterns in opioid vulnerability: investigating the links between disability, prescription opioids and opioid-related mortality", *BMC Public Health* 25, no. 1 (2025): 1759 [pdf]
- H. Fernando**, A. Spannaus, A. Kapadia, "A Geospatial Study of Opioid Overdose Risk in Diverse Regions", *Conference Proceedings for International Population Data Linkage Conference*, vol. 9, no. 5, 2024 [pdf]

Packages

- Spatial - TDA

Awards

- Merit Award for M.S. in Data Science
- Half Colors, Colors Awarding Ceremony 2017, University of Peradeniya, Sri Lanka

Presentations

- Unmeasured Spatial Risk in Disease Modeling: A Topological Data Analysis Approach, UT-Oak Ridge Innovation Institute's Second Annual Research Symposium, Knoxville, TN, Feb 2025
- A Geospatial Study of Opioid Overdose Risk in Washington D.C. Metropolitan Area, Interdisciplinary Association for Population Health Science Conference, St. Louis, MO, Sep 2024
- A Geospatial Study of Opioid Overdose Risk in Diverse Regions, International Population Data Linkage Network Conference, Chicago, IL, Sep 2024

Technologies

Languages: C, Java, Python, R

Technologies: RStudio, MINITAB, SPSS. Weka, Spark, Google Cloud IoT, Arduino